

AM6513 - ASSIGNMENT 3

DUE ON 28/08/2026

1. Make use of the Polynomial expression for the following functions: (a) $\frac{\partial \phi}{\partial x}$; (b) $\frac{\partial^2 \phi}{\partial x^2}$ and derive a general expression by (i) CD; (ii) FD; (iii) BD. Obtain the expression for non-uniform spacing and for two different orders of accuracy. Write a C / C++ / MatLab program to demonstrate the same. (The code should be as generic as possible).

2. In the class, we explained Central Compact Finite Difference Scheme (CCFD). Derive Upwind Compact Finite Difference Scheme (UCFD). Prepare notes and presentation. Share the literature from where you have understood this. 2. Develop a code implementing any one of Compact Scheme. Demonstrate the performance. Also compare conventional FD performance. Take a problem with analytical solution for comparison purpose.

It is your work. Send the report with code, code snippet, input data used to get results, obtained results and comparison performance. Code can be written in C++, Matlab or Python. Keep this solver safe. For future assignment, this code will be extended/augmented.

AM6513 - ASSIGNMENT 1

DUE ON - 19.08.2013

1. Make use of the Polynomial expression for the following functions: (a) $\frac{\partial \phi}{\partial x}$; (b) $\frac{\partial^2 \phi}{\partial x^2}$ and derive a general expression by (i) CD; (ii) FD; (iii) BD. Obtain the expression for non-uniform spacing and for 2 different orders of accuracy. Write a C / C++ / MatLab program to demonstrate. (The program should be as general as possible).

2. Obtain Compact FD for uniform spacing by central differencing and (forward/backward) upwind differencing for $\frac{\partial^2 \phi}{\partial x^2}$, for two different orders of accuracy. Tabulate and compare the different values of coefficients, different orders of accuracy and the truncation error.